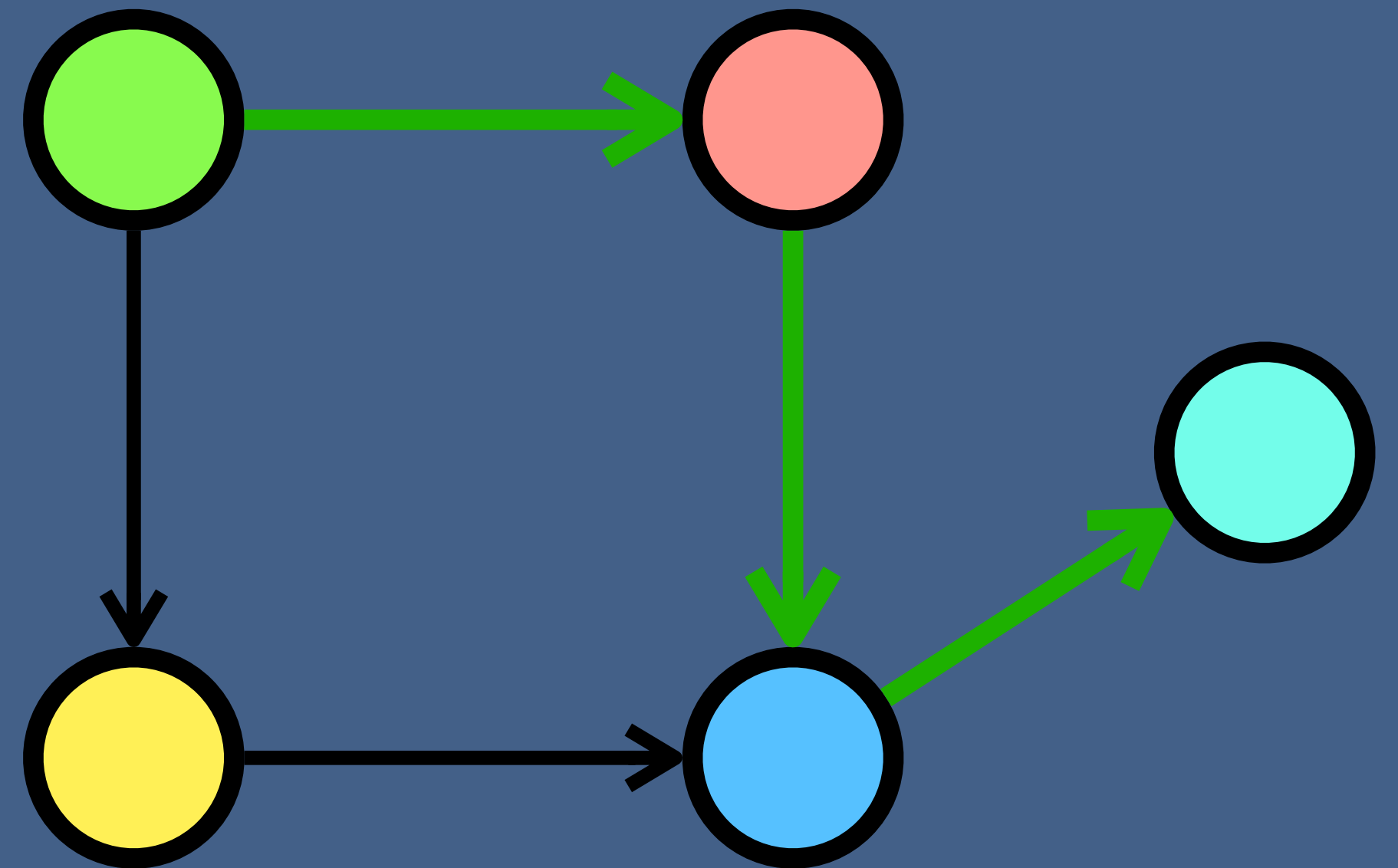


# All Pairs Shortest Path Problem (APSP)

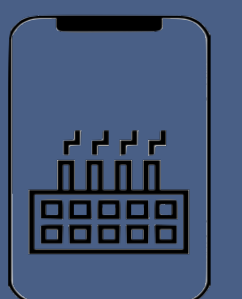
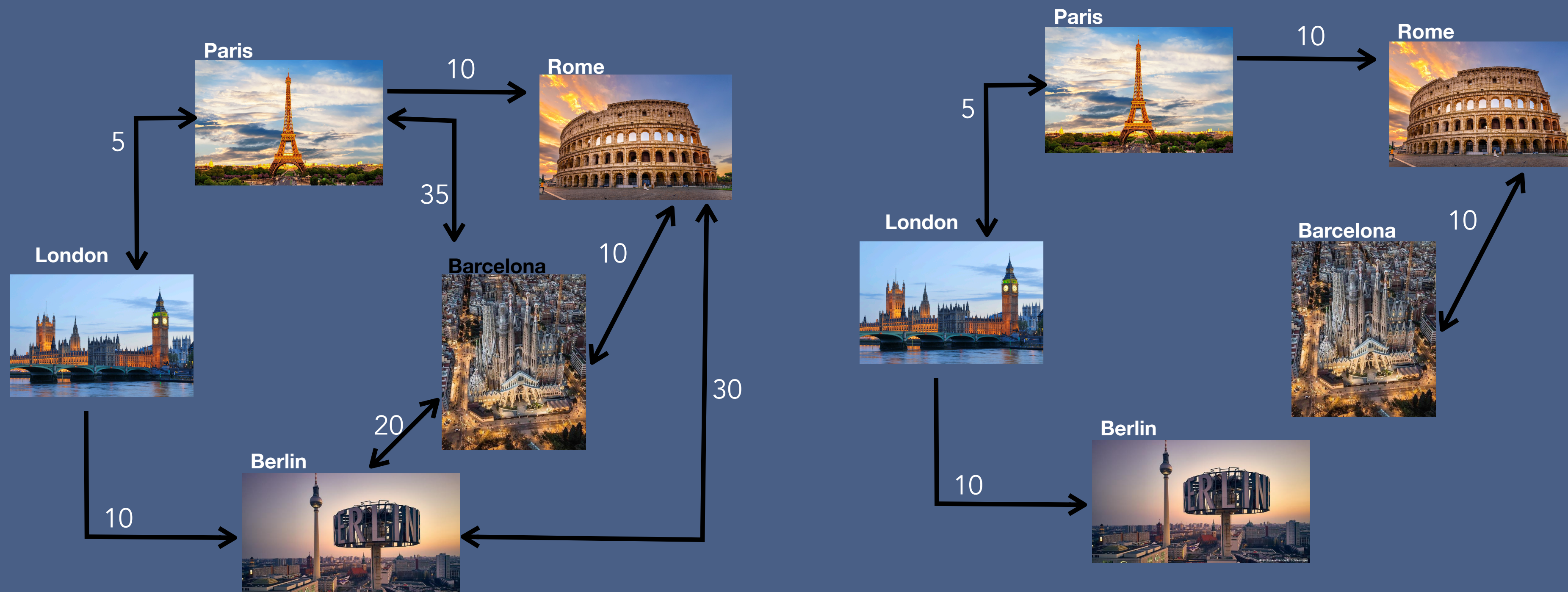


# All Pair Shortest Path Problem

## What is single source shortest path?

A single source problem is about finding a path between a given vertex (called source) to all other vertices in a graph such that the total distance between them (source and destination) is minimum.

- Five offices in different cities.
- Travel costs between these cities are known.
- Find the cheapest way from head office to branches in different cities



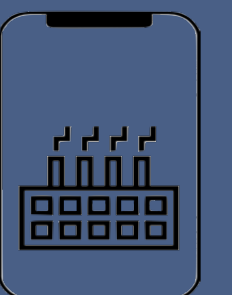
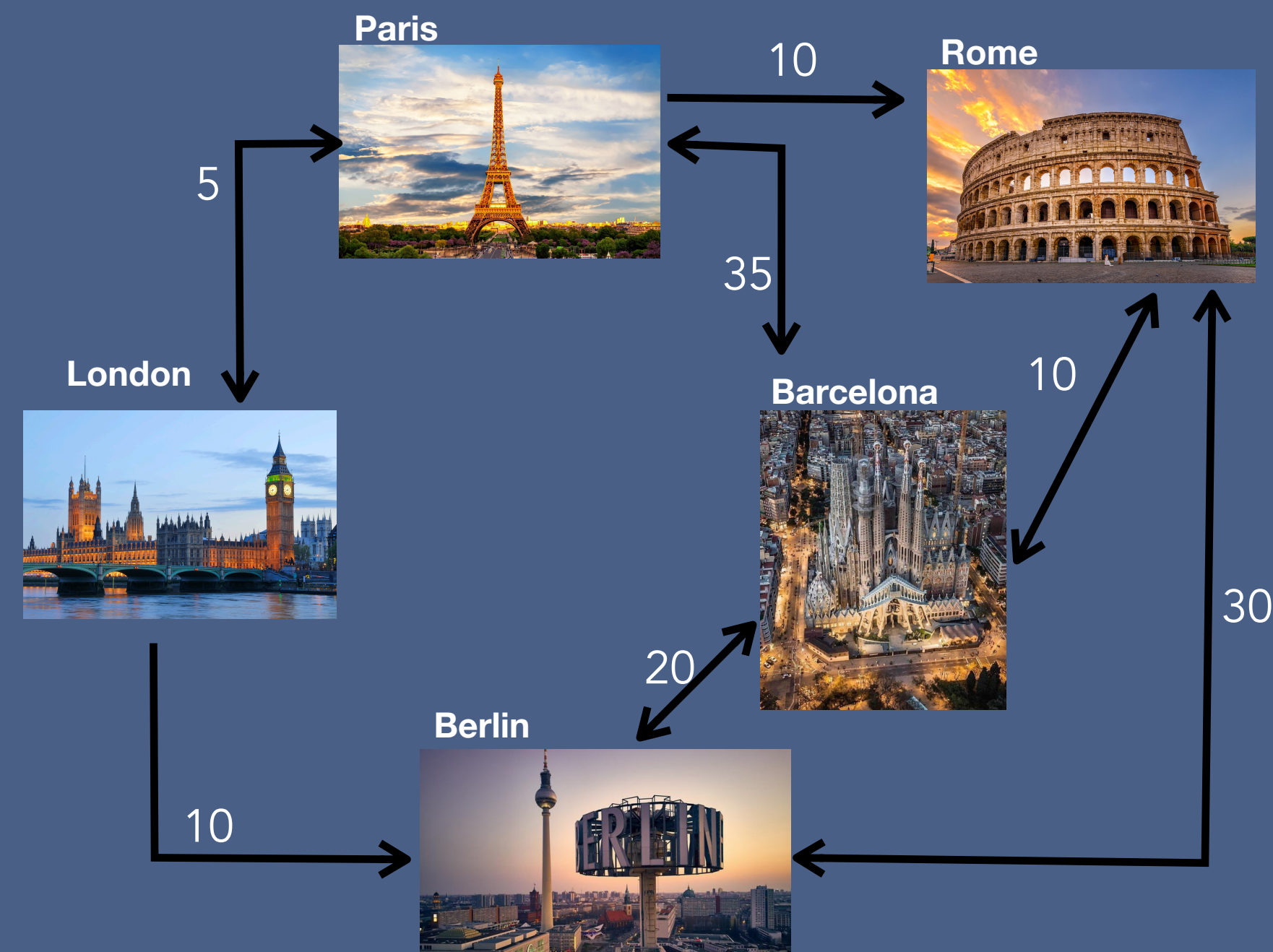


# All Pair Shortest Path Problem

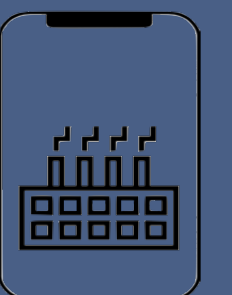
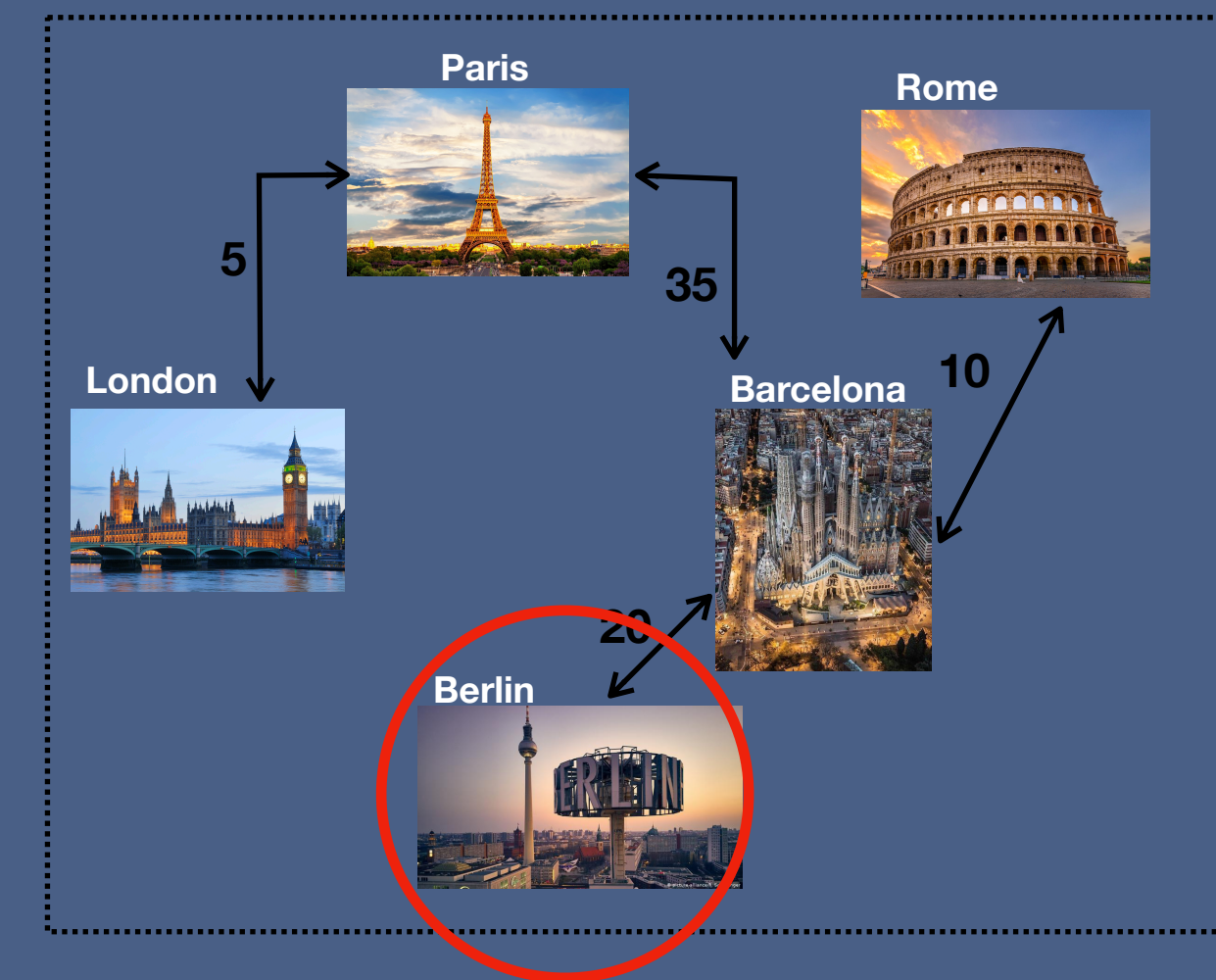
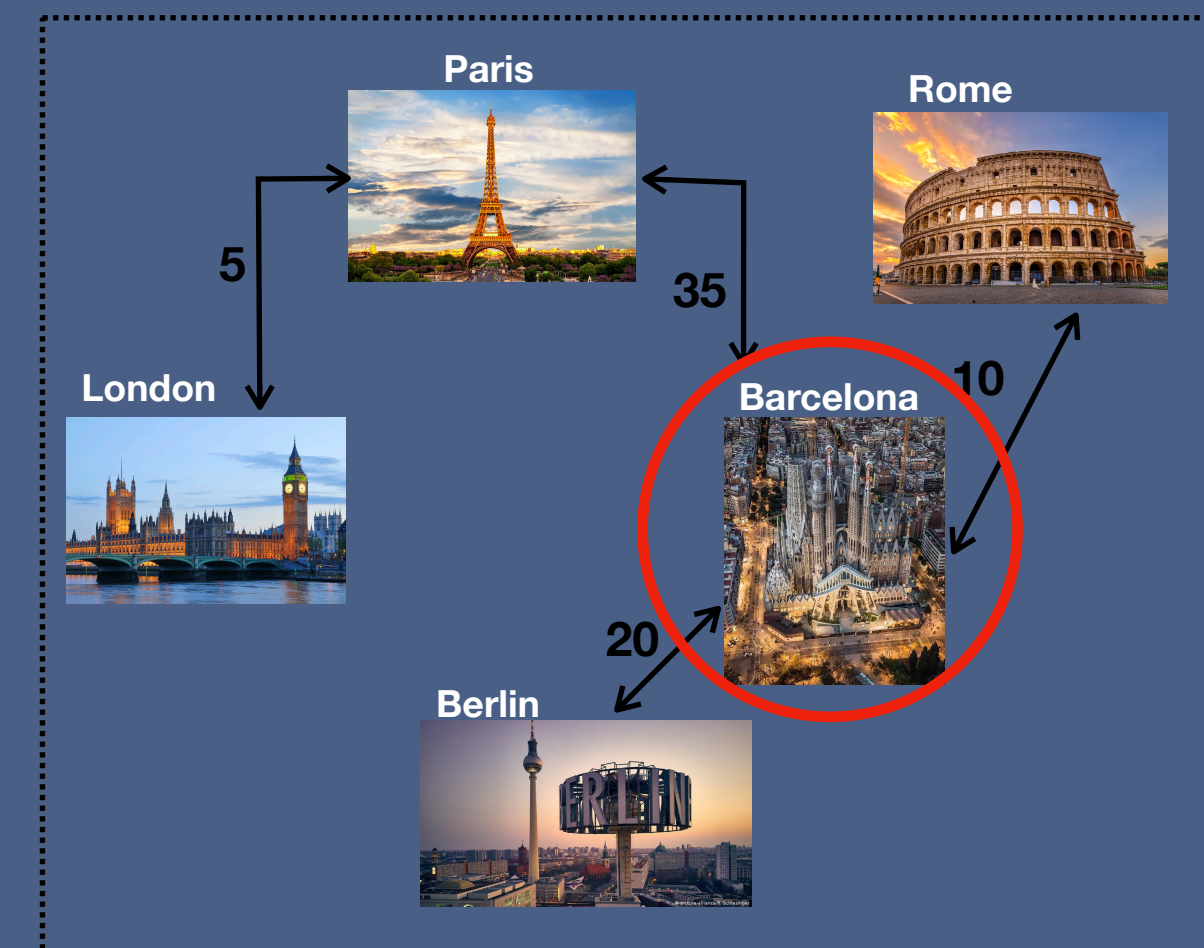
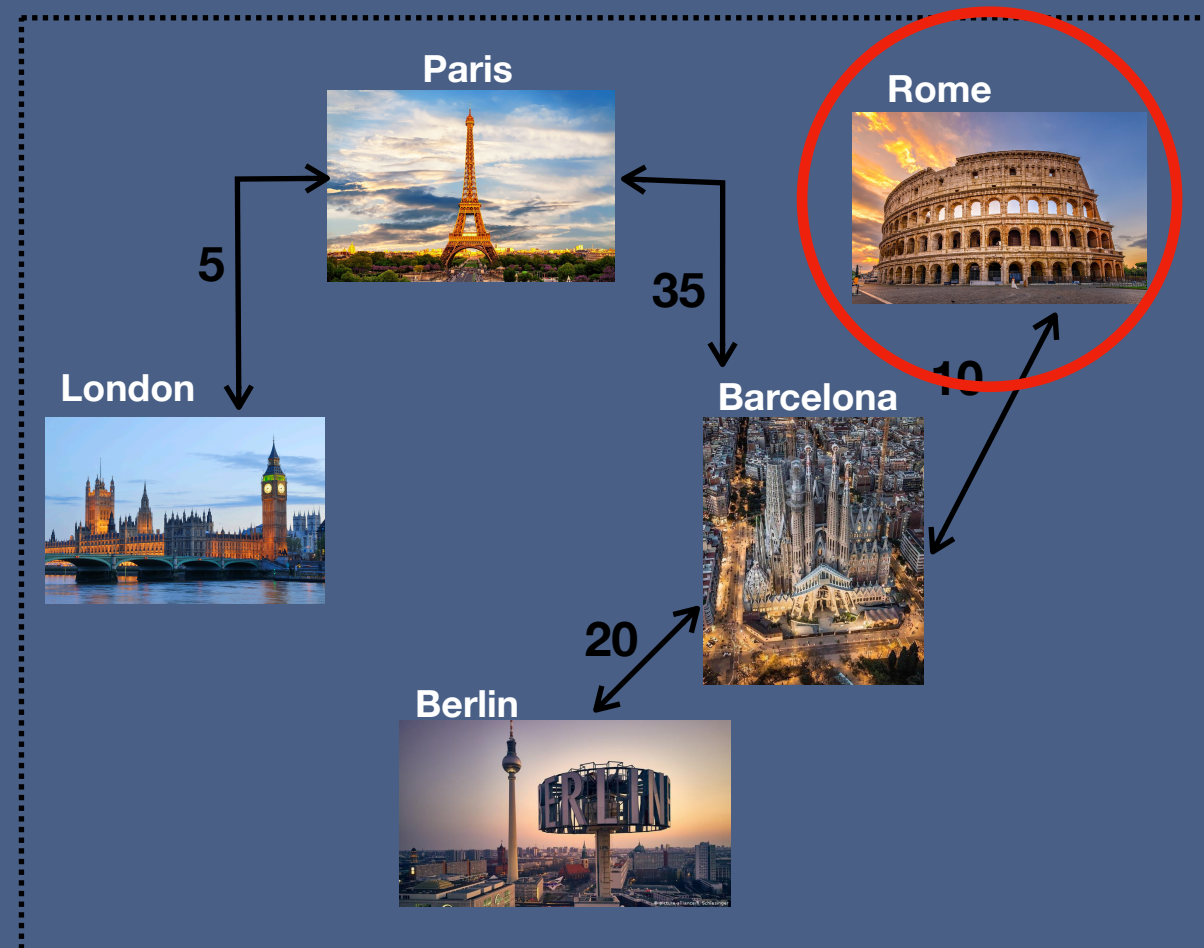
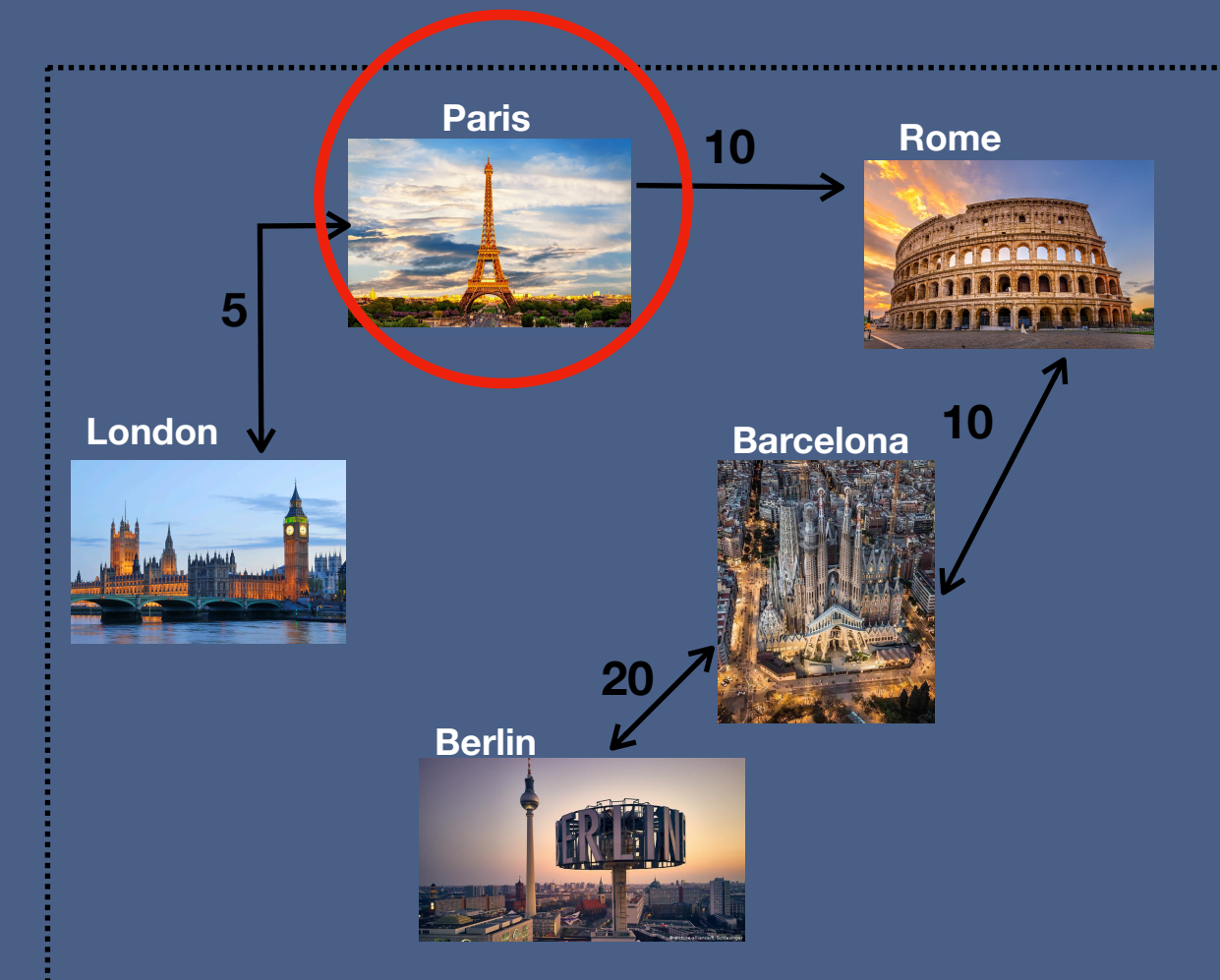
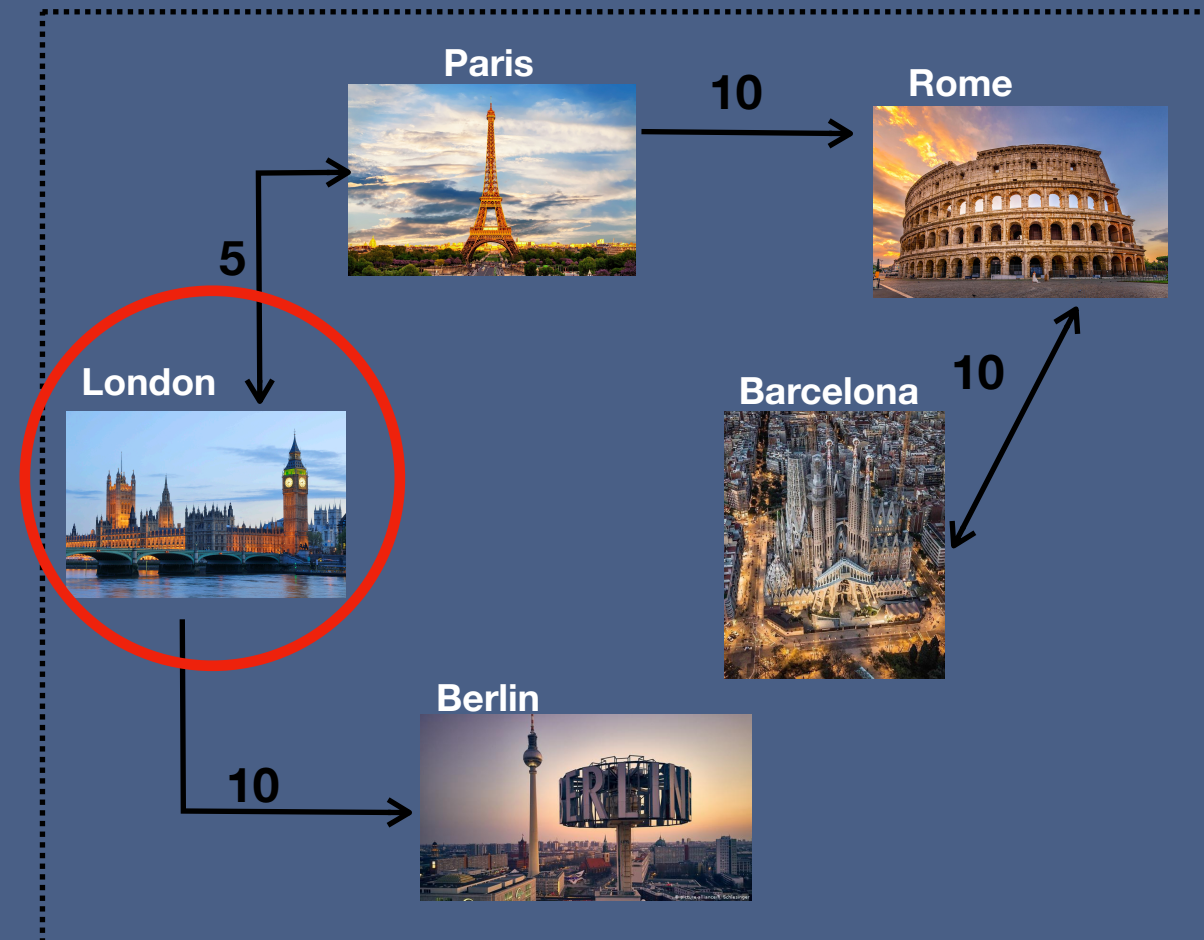
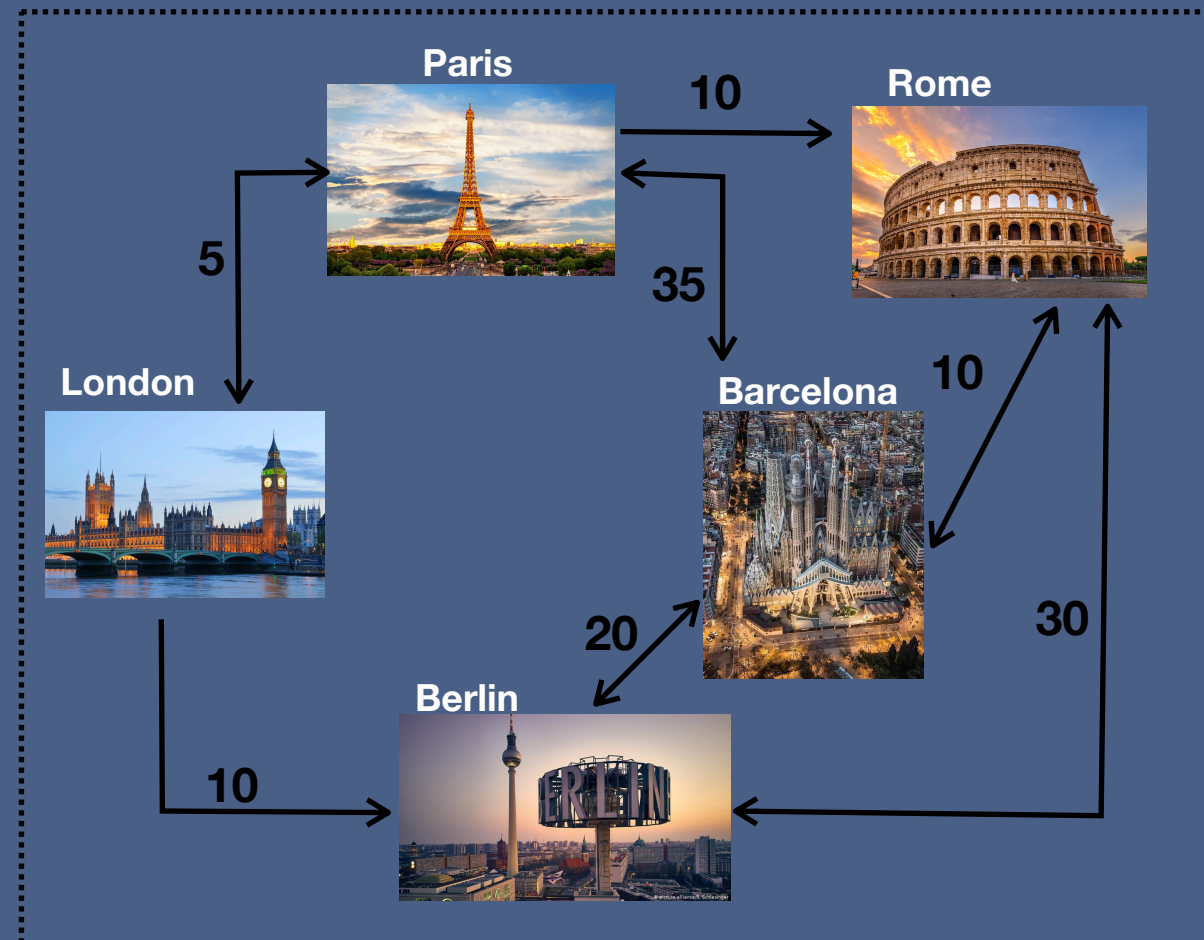
All pair shortest path problem is about finding a path between **every vertex** to all other vertices in a graph such that the total distance between them (source and destination) is minimum.

The problem:

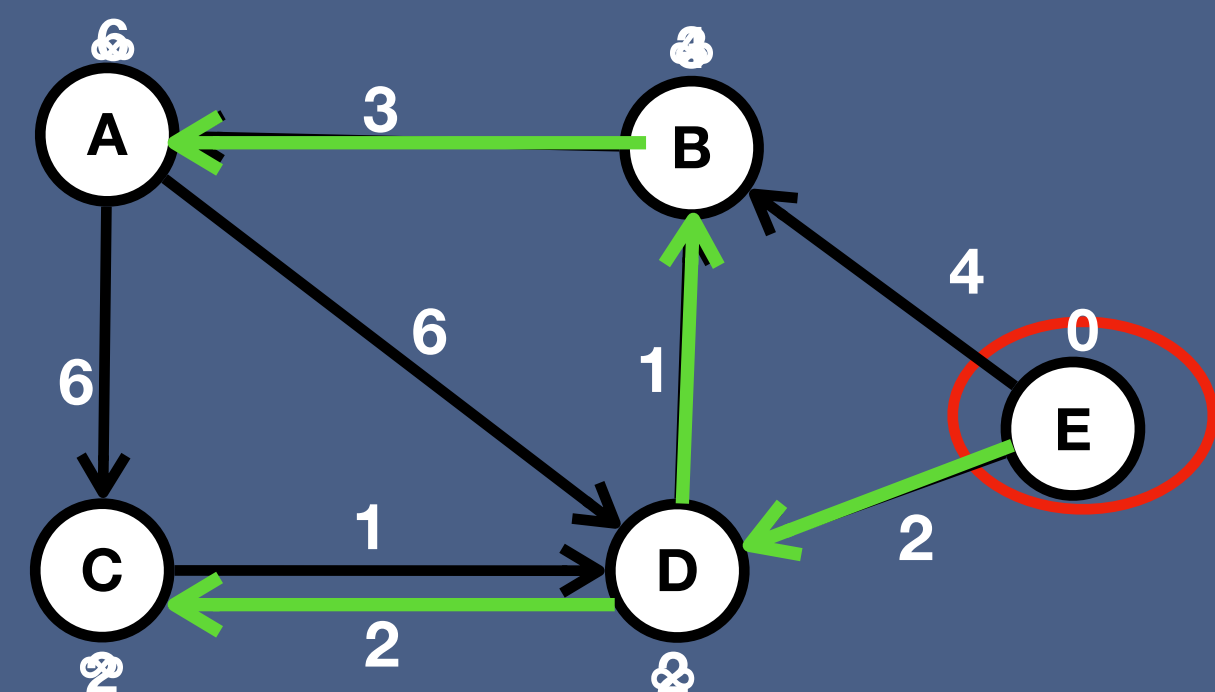
- Five offices in different cities.
- Travel costs between these cities are known.
- Find the cheapest way to reach each office from **every other office**



# All Pair Shortest Path Problem

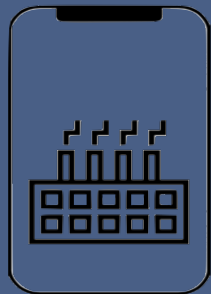


# Dry Run for All Pair Shortest Path Problem



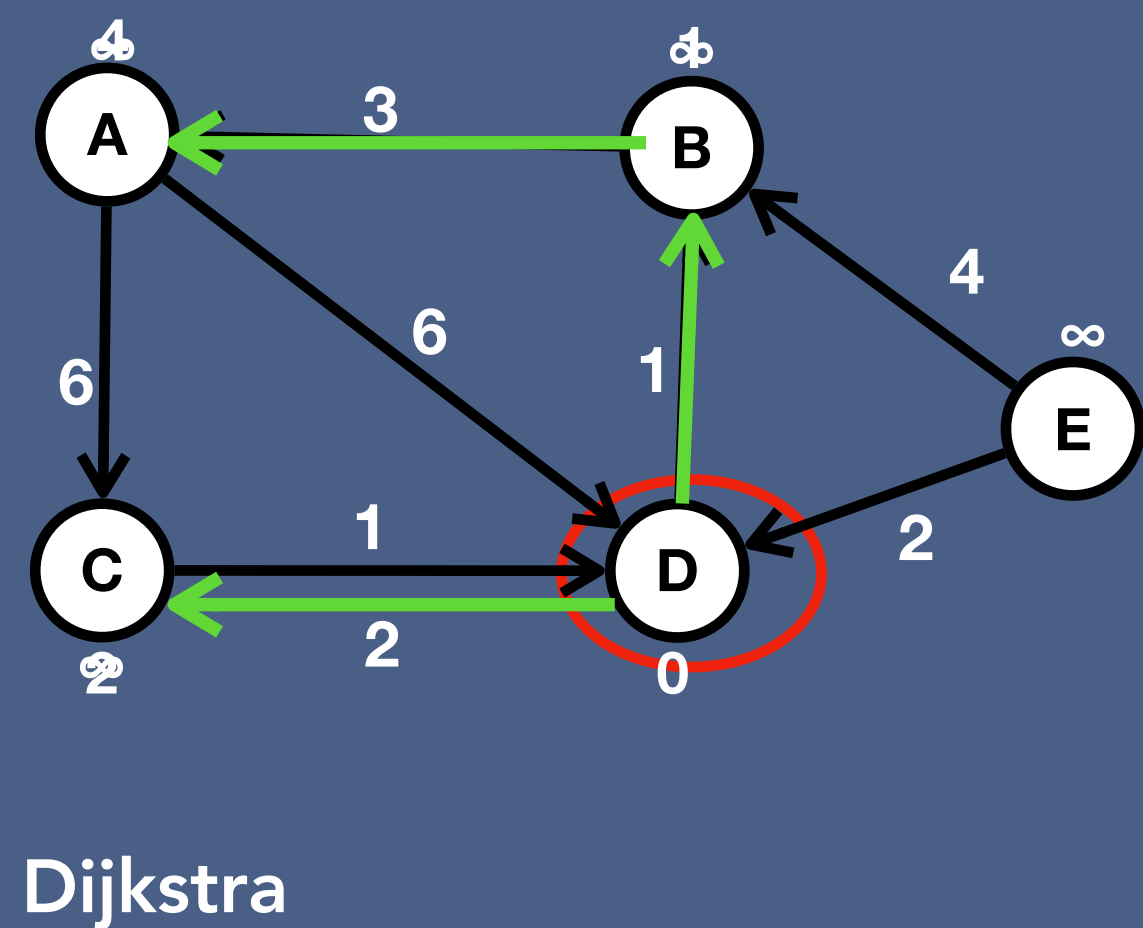
Dijkstra

| Source Vertex E | Path       |
|-----------------|------------|
| A               | E->D->B->A |
| B               | E->D->B    |
| C               | E->D->C    |
| D               | E->D       |
| E               | -          |





# Dry Run for All Pair Shortest Path Problem



| Source Vertex E | Path       |
|-----------------|------------|
| A               | E->D->B->A |
| B               | E->D->B    |
| C               | E->D->C    |
| D               | E->D       |
| E               | -          |

| Source Vertex D | Path    |
|-----------------|---------|
| A               | D->B->A |
| B               | D->B    |
| C               | D->C    |
| D               | -       |
| E               | N/A     |

